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Source: *Journal of Animal Ethics*, Vol. 4, No. 1 (Spring 2014), pp. 74-94

Published by: [University of Illinois Press](#) in partnership with the [Ferrater Mora Oxford Centre for Animal Ethics](#)

Stable URL: <http://www.jstor.org/stable/10.5406/janimaethics.4.1.0074>

Accessed: 19-09-2015 19:39 UTC

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The Morality of the Reptile “Pet” Trade

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The trade in, and private keeping of, reptiles as “pets” raises several ethical concerns regarding animal welfare (associated with handling, storage, transportation, intensive captive breeding, captivity stress, injury, disease, and high premature mortality); public health and safety (associated with zoonotic disease and animal-linked injuries); species conservation and environmental degradation (associated with wild capture); and ecological alteration (associated with invasive alien species). Also, many captive reptiles are fed other animals, raising broader ethical questions. Misperceptions about reptiles by proponents of their captivity mean that these animals are subject to conditions that would likely be considered unacceptable for dogs or cats.

KEY WORDS: reptile, pet, trade, ethics, premature mortality, welfare, stress, disease, environment, zoonoses

INTRODUCTION

Reptile trading and keeping has undergone several cyclic phases in popularity, with regard to the species and types that are marketed and the way in which they are sourced, sold, and maintained. Unknown tens of millions of individual animals and thousands of species are “wild-caught” or captive-bred annually to supply the exotic “pet” trade. According to Karesh, Cook, Gilbert, and Newcomb (2007), global “wildlife” trade for pets involves 350 million live animals a year, many of which are for “pets.”

About 25% of the “wildlife” trade is thought to be illegal (Karesh et al., 2007). Although some regulations are being increasingly adopted, these are mostly trade-permissive, with illegal and legal trade often sharing routes. Reptiles are a major component of this global trade. Data on the volume of the global reptile “pet” trade are unknown, and regional or country-based figures are also incomplete because few countries compile reliable data on reptile trade and keeping. However, annually perhaps 2 million live reptiles are imported into the United States, and 9 million are exported (HSUS, 2001, 2009). The United States is known to produce 3–4 million baby “pet” turtles annually, mostly for export (Warwick, 1997). In Rosenthal’s (2012) study, over 85% of individuals and over 90% of shipments

were recorded as “wild-caught.” UNEP/WCMC (2009) indicated that between 1997 and 2007, 12.3 million live reptiles (which largely involves “pet” animals) on CITES Appendix II were imported into the European Union—which represents only part of overall trade. The European Union is a very large market for “pet” reptiles. According to trade statistics, between 2005 and 2007 the European Union imported 6.7 million live reptiles (ENDCAP, 2012). In many countries locally caught reptiles are also kept. For example, in the United States historical estimates indicate that between 3 million and 5 million reptiles were “wild-caught” from Louisiana alone for the domestic market in the 1990s (Reaser, Clarke, & Meyers, 2008), and many believe that the trade has increased significantly since then. Reasonable estimates suggest that in the United States approximately 13 million reptiles are kept in homes (American Pet Products Association, 2011), and in the United Kingdom 700,000–800,000 (PFMA, 2012). These figures exclude the substantial numbers of reptiles who are either “wild-caught” or captive-bred for trade and who do not survive to reach private homes.

Reptile “sourcing” for the “pet” trade raises serious concern because whether they are “wild-caught” or captive-bred, securing these animals results in harm. For example, “wild capture” frequently involves injurious and stressful physical handling (resulting in broken limbs and tails and, where tortoises and turtles are concerned, crushed shells); local transport is often conducted using sacks or boxes on motorcycles over long distances (incurring more stress and damage); and storage frequently incurs insanitary holding sites with poor husbandry—all of which frequently leads to significant stress, suffering, and death (Warwick, 1997; 2001; Altherr & Freyer, 2001; Auliya, 2003; Laidlaw, 2005). Captive-bred individuals avoid some of these traumas but face intensive rearing conditions. Regardless of sourcing, reptiles are next subjected to packaging and transport in cramped, often insanitary, and undoubtedly fear-inducing conditions for long periods (Warwick, 1997; 2001; Altherr & Freyer, 2001; Auliya, 2003; Laidlaw, 2005). Arrival in recipient countries commences a new chain of challenges for reptiles, where typically poor husbandry, stress, and disease will culminate in the death of nondomesticated animals who have been transposed from their natural ecological niches to the unnatural world of the living curio.

In the reptile-keeping world, although the terms “pet,” “specimen,” and (to a lesser extent) “companion” are variously interchanged, here the term “pet” is of choice because it is the simplest way to describe a nonagricultural animal who is arguably not a genuine companion—reptiles have little to no normal behavioral interaction with humans and thus are not true “sharers” of our lives. A historical and, in particular, recent raft of peer-reviewed articles and reports reveals that practices involving both the trade and the private keeping of reptiles as “pets” are laden with issues that raise ethical concerns. In brief these are as follows:

Animal welfare—stress, morbidity, and premature mortality accompany many if not most animals at all points in the trade and keeping chain (see, e.g., Franke & Telecky, 2001; Warwick, 1997, 2001; Laidlaw, 2005; ENDCAP, 2012; Toland, Warwick, & Arena, 2012).

Public health and safety—zoonotic disease is now recognized as an emerging problem and a significant and major public health hazard. Injuries from exotic “pets” are a related concern (see, e.g., Brown, 2004; Chomel, Belotto, & Meslin, 2007; Karesh, Cook, Bennett, & Newcomb, 2005; Jones et al., 2008; Praud & Moutou, 2010; Abbott, Ni, & Janda, 2012; Hale et al., 2012; Warwick, Arena, Steedman, & Jessop, 2012).

Species conservation and ecological alteration—species conservation threats arise as a result of taking animals from natural populations, and ecological alteration arises from incidental releases of unwanted invasive animals into novel habitats (see, e.g., Auliya, 2003; Krauss et al., 2003; Bomford, Kraus, Barry, & Lawrence, 2009; Kark, Solarz, Chiron, Clergeau, & Shirley, 2009; Shine et al., 2010; Vilà et al., 2010; Henderson & Bomford, 2011; Langton, Atkins, & Herbert, 2011).

Also, offering live foods is a frequent component of the business of selling and keeping many reptiles, and this raises additional questions.

Reptile traders and keepers maintain that there are important side benefits to their activities, including species conservation, arising from financial incentives for local animal collectors to preserve indigenous species; economic value from wholesale and retail commerce; personal satisfaction; formative propagation of professional biologists; and the encouragement of public interest in nature.

HUMAN PERCEPTIONS OF REPTILES

Modern-day reptiles have a long history that, according to group, can be measured at least in the tens of millions of years. Part of that success is attributable to their innate (unlearned) behavioral and psychological profiles. Highly precocious, they are born ready and able to pursue prey, avoid predators, find shelter, and so on. Innateness does not imply an inability to learn or to think. Rather, it means that reptiles are strongly hard-wired to their evolved lives in the natural world.

Some claim that certain species of reptile have become domesticated through multiple generations of captive breeding and artificial selection. However, there is no evidence of true domestication in reptiles. Many desensitize to novel threats as they do to a swaying branch in nature and thus may seem “tame.” This kind of behavioral quiescence, though, is not an indicator of adaptation to captivity (Warwick, 1990; Warwick, 1995). Certain individuals of certain species—for example, bearded dragons (*Pogona vitticeps*) and royal pythons (*Python regius*)—have been selected and bred to enhance their natural tendency toward docility. Species that adapt well to domestication possess particular “pre-adaptive traits,” such as affiliative behavior toward people (Price, 1984; Cameron-Beaumont, Lowe, & Bradshaw, 2001), and reptiles generally do not possess these traits (Warwick, 1995; Warwick, Arena, Jessop, Lindley, & Steedman, 2013). The evidence demonstrating the lack of adaptability (and domestication) of even the most selectively bred species (including bearded dragons) is freely observable in their behaviors in artificial conditions (Warwick, 1995; Warwick et al., 2013). Basically, captive-bred lizards

show the same captivity stress-related signs as their free-living counterparts when those conspecifics are also caged (Warwick, 1995).

Reptiles need nature, and their evolved constitution frequently conflicts with artificial conditions, of which a minimalist vivarium in a bedroom is a common example. So although in the natural world innateness is a strategic advantage, in captivity it becomes a disadvantage. However, largely because of their relatively slow metabolism, reptiles often tolerate poor captive conditions and disease for long periods, giving a false perception that they are "thriving" while actually they may be languishing under stress (Warwick, 1995; Laidlaw, 2005).

Being ectothermic means reptiles are particularly dependent on environmental temperatures to regulate their thermal needs. Popular terminology describes this feature as "cold-blooded," which is actually a somewhat neutral designation, but one that also gets used to convey stoicism or insensitivity. In addition, reptiles are often seen as belonging to the category of "non-cute," and thus gaining sympathy or empathy for them is relatively difficult (Estren, 2012).

One might presume that false perceptions of reptiles reside entirely with those who sport dislike or disinterest and that the "fans" of the so-called cold-blooded animals might be universally able to set themselves apart from those who are reptile-naïve. However, if one considers a few of the common justifications by reptile traders and keepers for confining reptiles in minimalist (or in some cases possibly "naturalistic") cages, the following supposed facts emerge to underpin the claims: Reptiles are "not social, and therefore have reduced needs"; they are "unintelligent, and do not notice their environment"; they are "agoraphobic, and do not need space"; and they "need only food, water, 'proper' temperature, lighting and humidity to lead healthy lives" (Warwick, 1997; 2001). In addition, reptiles who are "good feeders," "good growers," and "good breeders" are viewed as showing all the necessary hallmarks of good well-being.

Notwithstanding that even these very basic goals for "good well-being" are often not achieved, all of these, along with other perceptions, are either weak at best or quite simply wrong. Many reptiles are social and rival mammals for measurable intelligence; all those studied are highly aware of their environments; agoraphobia is a human anxiety disorder and not known in reptiles; range studies reveal both large and small reptiles to be highly active and to need plenty of space; and environmental, physiological, and behavioral needs are often poorly understood (even for "common" species), meaning that "proper" feeding is not a valid concept (Warwick, 1995; Warwick & Steedman, 1995; Arena & Warwick, 1995).

A result of these erroneous perceptions and presumptions is that most captive reptiles will suffer drawn-out stress, morbidity, and premature mortality because they are nondomesticated animals confined to diminutive enclosures wherein many (particularly snakes) will not even be able to extend their bodies to full length. Although false perceptions by the general public may cause reptiles to be shunned or ignored, it may be the misperceptions of their admirers that result in some of the greatest harm.

ANIMAL WELFARE

The best evidence confirms that reptilian sentience and sensitivity are comparable with that of other animals, including humans (Warwick, 1990; Gillingham, 1995; Warwick, Frye, & Murphy, 1995; Warwick, 1995; Warwick et al., 2013). Whether one examines pain and aspects of anatomy and physiology for neurological structure, reception, and transmission; emotion through physiology and behavior; or psychological factors relating to stress, physiology, arousal, quiescence, and comfort, one can quickly recognize the commonalities of consciousness. In fact, early evidence of these characteristics has existed for at least three decades.

Important differences exist between domesticated companion animals and reptiles in that whereas domesticated animals—for example, dogs and cats—generally experience relatively liberal home- and life-sharing associations with humans, reptiles are typically caged for life. This not only takes evolved life management—survival programming—out of the control of reptiles, but also places it firmly in the hands of human beings who, with very little actual knowledge, take on a highly selective role as god and gatekeeper.

If this does not make the point strongly enough, then perhaps consider the responsibilities, demands, and knowledge involved in having and bringing up a child. Modern life provides us with ready access to more or less everything we need to bring a child into the world and send the child on his or her way. Doctors, nurses, surgeons, midwives, psychologists, pharmacists, schoolteachers, ambulance drivers, nutritionists, family, and others are on tap to advise on topics from getting pregnant to how to deal with a departing daughter and everything in between. All these professionals are well qualified, welfare-directed, and readily accessible. But despite the wealth of expert assistance available on “child-keeping,” perhaps the most important factor not on this list is that having a child is the result of “what comes naturally.” Nature provided us with a suite of physiological, anatomical, psychological, and behavioral attributes that would probably equip us to handle the child-care job on our own from start to finish. Yet despite both nature and nurture seeing us through the raising of our offspring, would we describe the process as easy? Probably not!

Compare all that with keeping a reptile. Not only is the reptile not human, but it also is not even a mammal. It belongs to a distant biological class and has significantly different physiological, anatomical, psychological, and behavioral attributes. It is usually from a different part of the world and inhabits an ecosystem about which we know very little. Similarly, we know very little about the natural biological needs of reptiles, and what we do know appears to indicate that we do not, and probably cannot, provide for them. Finally, lay all these problems in the control of the reptile trade and keeping fraternity, which frequently harbors misperceptions about reptile needs, and the result is nature denatured at the very least or catastrophic failure of welfare, which is the typical consequence.

Stress, Morbidity, and Premature Mortality

At least 30 captivity stress-related behaviors and many more diseases and injuries are regularly observable in most kept reptiles (Frye, 1991; Warwick, 1995; Warwick et al., 2013). Many diseases and injuries are identifiable from reptile behavior in the same way that a limping dog conveys leg damage. However, some behaviors are signs of captivity stress, such as hyperactivity and interaction with transparent boundaries, both of which involve persistent attempts at escape, and hypoactivity, which involves efforts to biologically “shut down” from a poor environment. What is more, these problematic behaviors are extremely common and can themselves become self-compounding and physically injurious. Despite being regularly observed, many of these signs are simply ignored by keepers as irrelevant, a disregard that marks a failure in understanding of important behaviors.

Stress, morbidity, and mortality linked to reptile trading and keeping occurs at a variety of hubs. Mortality associated with “wild capture” is unclear, but observations range from 5% to 100% according to species, and captive breeding frequently involves a mortality rate of 5–25% (Toland, Warwick, Arena, & Steedman, n.d.). At wholesalers, premature mortality has been found to be approximately 70% during a six-week period, which is also the reported industry standard (Toland et al., 2012). Shipping or transport mortalities can be measured within the margins of 1–100% according to species and situation (Toland et al., unpublished).

It should be borne in mind that although most shipments may be in the 1% mortality range, those deaths occur in relatively short (e.g., 12– to 24-hour) periods and thus represent a large number of animals given that globally numerous millions of animals are involved. A 1% transport mortality rate applied to human travel would represent approximately four people dying on a single long-haul flight from London, England, to Perth, Australia. Losses at retailers are thought to be approximately 12% per week (Toland et al., n.d.).

A study of reptiles (turtles, tortoises, lizards, and snakes) in the United Kingdom found that the premature mortality rate was 75% in their first year in the home and 81% annually if one includes wholesale and retail losses (Toland et al., 2012). By comparison, a study of life span for domesticated dogs in the United Kingdom indicated an average of over 11 years (Mitchell, 1999). These studies used different assessments (average longevity versus premature mortality), but the net result is that most dogs achieve natural longevity whereas most reptiles do not.

Survival and Mortality in Nature Versus Captivity

A commonly presented counter argument to trade and keeping mortality is that captive reptiles live longer than those in nature and that high mortality in captivity is no worse than natural attrition in the “wild”—that is, the attitude is that a loss is a loss. So if nature itself could not better the attrition rate, then should humans, as animal managers, be held to higher standards?

First, it is by no means clear whether animals in nature generally live shorter lives than captive animals, or indeed what proportion do and do not achieve their natural potential longevity. There is some evidence to show that some individual reptiles do live long captive lives. However, it is untenable to claim that these examples are unique to captivity, when there are few or no comparable opportunities to examine longevity of individuals in the “wild”—which may also reveal long individual life spans. There is a general paucity of data on animal life spans in nature, and thus the argument is not evidence-based. Even where the argument may be true, for a few animals anyway, the issue of quality of life requires consideration. An animal in nature is doing (or even dying while doing) what he or she has evolved and is adapted to do, whereas, by contrast, a captive animal cannot achieve this fundamental prerequisite of life.

Second, once in our care, causes of natural attrition are largely altered or omitted, and animal well-being is transferred to our responsibility. Even in captivity, longevity remains a natural goal. Also, when early-life mortality in nature is high, these losses typically occur within an ecological context and assist in the survival of other animals, which is unlike captive animal attrition, where there are few or no relevant direct beneficiaries of death. Nature can no longer be blamed for an animal’s health, welfare, morbidity, or mortality—the animals’ suffering and premature death are essentially in *our* hands. Therefore, it is plainly arguable that comparing morbidity and mortality in nature versus captivity is a mismatch.

Third, as humans we regard *potential* longevity as a measure of normality and premature mortality as a tragic event that should have been averted wherever possible. In most parts of the world, we do not measure acceptable human life spans by what may be considered typical in global regions of high premature mortality. Similarly, if at least three out of four domestic dogs died within their first year in the home due to maladaptation and human negligence (as do reptiles), then this would very likely be considered unacceptable. Indeed, many people might regard longevity of even several years among domestic dogs to constitute tragic deaths. Whether it is a human who dies at 40 years or a dog who dies at 4 years, both are considered lamentable deaths that should not have happened.

However, is it being selective to rule out natural attrition as a comparison for captive attrition while at the same time accepting that potential long life spans, as experienced by many animals in nature, should form the basis of what are acceptable life spans in captivity? No, it is not. Longevity is a biological potential and a marker of success, indicating, for example, an organism’s ability to overcome adversity—its fitness. In nature a short life span may not indicate any sort of failure because a brief, productive life within a natural context may be highly beneficial for the individual, the individual’s species, and his or her environment. By comparison, there are no benefits for either a long or short stressful life in captivity.

Ethical consistency requires that the view of successful longevity we use for ourselves and for dogs should also be the marker for captive reptiles. Unfortunately, the evidence suggests that reptile traders and keepers, along with legislators and regulators, are currently prepared to apply altered and inferior standards for reptiles.

Live Food and Feeding

Sacrificing one live animal to a preferred species is very much part of reptile keeping, with animals who might to another person be a favored companion ending up as prey. Fishes, frogs, and birds are fed to snakes; snakes are fed to cannibalistic snakes; lizards are fed to snakes; and rodents (rats, mice, and their babies) and many invertebrates are fed to just about any reptile. So even species (such as gecko lizards) that one might intuitively think are "safe" in the appreciative hands of a reptile keeper are sometimes freely immolated as a meal for a more "interesting" animal such as a tree snake.

Although certain preferences are also made for other species, such as giving cats canned food made from fishes, this is probably some way from routinely feeding a cat live trout—let alone feeding a cat another live cat! Also, cats are already domesticated—it is a human responsibility to care for existing cats, whereas reptiles remain nondomesticated animals who do not need us, and thus we have no obligation to keep them.

Indeed, there exists a substantial allied industry of intensively produced live food (typically invertebrates and rodents) to keep captive reptiles nourished. The treatment of these so-called "food animals" is often concerning because during their short captive lives they are regarded as—and manifestly become—fodder. Lee et al. (2007) described conditions of rodent suppliers in Texas where stocking density was 1,800 rats and mice per 4.6 m x 4.6 m room, at 12 rodents per (45 cm x 41 cm x 20 cm) tray and 154 trays per room. The scale of the global trade in rodents as food animals is unknown. However, Lee et al.'s (2007) study of just four Texas-based suppliers found that between them they produced about 4.2 million rats and mice annually.

Approximately 1–2% of these rodents die, largely as a result of difficult births, overgrown teeth, and drowning in overfilled water trays during production. Lee et al. (2007) also noted that the rodent feeder industry has not been well regulated throughout the United States and recommended greater scrutiny.

Globally, rodents (typically rats and mice) for reptile food who are killed individually are commonly terminated by cranial percussion (striking the head of the animal or striking the animal against a hard object), which may cause rapid concussion and death or result in convulsions and death. Similarly, whole-body impacts involve placing a small rodent in a bag or sack and striking the sack against a table or other hard object. Cervical dislocation is also used on the smaller animals. Large numbers are often killed using carbon dioxide (Conlee, Stephens, Rowan, & King, 2005). However, CO₂ is an irritant gas, which induces desperate and painful suffocating sensations (Close et al., 1996a, 1996b; Conlee et al., 2005). When supplied live, and aside from the handling, packaging, and transport processes, the fate of a rodent will depend on what predator awaits. These issues make the point that both reptiles held for their keepers' pleasure and those animals who will become those reptiles' artificial food may all be subjected to unnecessary suffering as a result of a spurious hobby.

Large lizards are brutal attackers, using powerful jaws, strong curved teeth, and strong curved claws to damage and dismember prey. Snakes may attack by biting, constricting (causing death by asphyxiation and/or circulatory strangulation), or using poison. It is

particularly difficult to estimate the extent to which fear might impact on the general traumas experienced by prey animals, but it is likely not insignificant. For smaller predators (certain snakes and lizards), prey will frequently consist of invertebrates such as crickets, larvae, fishes, or frogs, and these are either chewed slightly or swallowed whole—and alive. There is now good evidence to suggest that invertebrates (including crustaceans, insects, arachnids, and others) are sensitive to pain and other aversive stimuli; thus, the welfare of these animals used as food correctly warrants concern (Smith, 1991; Elwood, 2011; Crook, 2013; Horvath, Angeletti, Nascetti, and Carere, 2013). Under natural conditions the act of a reptile or other predator subduing and consuming prey may be very different from what occurs when a living animal is merely cast into the domain of a hungry captive individual. In nature, long-evolved biological strategies probably exist that ameliorate the predator–prey relationship. For example, in free-living animals, antipredator behavior and physiology include exceedingly powerful stress- and pain-reducing responses such as endorphin release mechanisms that likely assist to minimize the trauma of attack (Warwick, 1995). In captivity, these normally involved or incidental mechanisms of nature may be circumvented by the artificiality of husbandry, to leave prey animals both behaviorally and physiologically deprived of essential strategic protections that make life and death more humane.

Reptiles are no less humane killers than other animals, and their evolved dietary needs and prey apprehension strategies, whether entirely humane or not, are nevertheless part of nature. In captivity, however, any inhumanities arise because of the human practice of choosing to keep animals in an unnatural situation where people become the middlemen between those who live and those who die, in this case not as part of natural survival, but as part of a hobby.

PUBLIC HEALTH

Around 200 zoonotic (animal-to-human) diseases exist, and at least 40 are associated with reptiles (Warwick, Arena, Steedman, & Jessop, 2012). Most notable among these is reptile-related salmonellosis, which accounts for between 1% and 5% of all salmonella infections or about 70,000 cases in the United States and 5,600 in the United Kingdom annually (Toland et al., 2012). It is estimated that 61% of human diseases may have a zoonotic origin (Karesh et al., 2005) and that 75% of global emerging human diseases are of animal origin (Brown, 2004). The selling and keeping of reptiles and other exotic animals has been likened to a Trojan horse of infection and infestation because people acquiring these animals are commonly unaware of the risks of introducing stealthy and infiltrative diseases right into the home (Warwick, 2006; Smith et al., 2012).

Although domesticated animals—for example, dogs and cats—are frequent carriers of zoonotic agents, most veterinarians and doctors are aware of the more common zoonoses and may quickly recognize many risks and advise accordingly. However, veterinarians encounter exotic “pets” (which harbor more diverse and novel pathogens) less frequently, so zoonotic disease may go unrecognized or misdiagnosed, and exotic

agents are frequently more persistent and severe than domestic ones. Combined, both the relatively low veterinary and medical awareness and the more diverse, and in some cases more robust reservoir of potential pathogenic agents, make reptiles a particularly worrisome risk around the house.

A notorious source of poor animal treatment and selling practices is reptile expos, breeders’ meetings, and shows—all of which are essentially rummage sale or car boot–like “pet” markets. A study of such European markets found that public health risks at markets were significant and “appear unresolvable given the format of the market environment” (Arena, Steedman, & Warwick, 2012; Warwick, Arena, & Steedman, 2012). Another recent academic assessment of 56 breeders and sellers at a reptile expo in the United States found that 17.4% of people stated they either had been ill with reptile-related salmonellosis or had known someone who had been ill (Rosenthal, 2012).

In the United States, in the 1960s and 1970s, an estimated 280,000 cases of “pet” turtle–related salmonellosis occurred annually, despite governmental efforts at educating people on prevention and control (Mermin et al., 2004). No measures worked until the trade in baby slider turtles was banned in 1975, and the following year saw a 77% reduction in cases of the disease (Mermin et al., 2004). Currently in the United States it is estimated that around 70,000 cases of reptile-related salmonellosis occur annually (Mermin et al., 2004).

Decades of formal and informal intervention and guidance on preventing reptile-related human salmonellosis have merely toyed with the problem. Abbott et al. (2012) conclude,

Unfortunately, despite recommendations to the US public beginning in the mid-1990s, regarding the potential risk for acquisition of salmonellae infections from exotic pets, the number of infections in the US caused by subspecies associated with these sources does not appear to be abating. (p. 638)

Much guidance has centered on hand-washing to reduce human infection risk after touching a reptile. However, this somewhat naïve caution overlooks the incidental transference and transmission of bacteria from hands to nearby surfaces, which then recontaminate hands, mouth, and gut (Warwick, Arena, Steedman, & Jessop, 2012). The only way for an individual to be safe and—to not present a risk to others—is to not handle reptiles at all nor touch those who handle them.

Modern science and politics have embraced the somewhat holistic philosophy that the fates of environment, animals, and peoples are all entwined—popularly labeled as the “one health” or “healthy animals, healthy people” message. Captured, caged, mishandled, and misunderstood reptiles get stressed. It is bad for them, but stressed animals also are more likely to shed infectious agents to others and to us. It is a sound message—stress in any link of this chain adversely affects the next.

There is no cure for “pet” reptile and other animal zoonoses so long as the capturing, breeding, selling, and keeping of these animals continues. Major commercial “producers” have tried and failed to eradicate even regular bacteria from, for example, turtles, and in fact this (as with other animals, such as ornamental fishes) has led to the development

of antibiotic-resistant bug strains (Diaz, Cooper, Cloeckaert, & Siebeling, 2006; Rose, Hill, Bermudez, & Miller-Morgan, 2013). As Akhtar (2012) explains, a “considerable amount of human suffering may be avoided if rather than asking ourselves how to thwart an epidemic once it has begun, we ask whether we can prevent an epidemic by treating animals differently.”

Diseases, including zoonotic forms, among reptiles and other nondomesticated animals are now fully established as significant and major threats to both human health and the health of animals who are farmed, and potentially major pandemics such as severe acute respiratory syndrome (SARS) and avian influenza have already risen in association with the exotic “pet” trade. It cannot be ignored and should not be downplayed that within the trade and keeping of reptiles and other nondomesticated animals, there likely exists a global reservoir of novel, durable, and invasive pathogens. The emergent strategies of these bugs may be latent, dormant, or insidious or spontaneous and aggressive. So many human and animal health problems relate directly to the way in which animals are sourced, transported, stored, treated, and kept. Some improvements consistent with commerce may be possible, but resolutions are not. As the nonchalance of exotic “pet” selling and keeping increases, and the trade-favoring laxity of government lumbers forward, so too do the risks of unleashing the next major AIDS-like pandemic or other biological catastrophe.

Many commentators focus on the joy or comfort people derive from “owning” an animal, be the animal reptilian or other. But when animals suffer and die as a result of such “ownership,” a perceptive mind would recognize exotic “pet” keeping as an indulgent and confounded pleasure. Overlooked, too, can be the stress, rumination, and depression that, for the sensitive caregiver, inevitably accompanies the bereavement of any animal. So significant can be the loss of an animal companion that today psychologists and special animal grief counselors are relatively common and often busy (Akhtar, 2012).

SPECIES AND ECOLOGICAL CONSERVATION

As mentioned elsewhere, the global market for reptiles is virtually unknown, which is in itself a disturbing fact. It is known that the reptile “pet” trade is very large indeed and probably expanding. Although captive breeding is significantly more common today, the trade continues to involve a major “wild-caught” component (Marano, Arguin, & Pappapionou, 2007). The effects of trade on species and ecologies are becoming increasingly known. For example, Schlaepfer, Hoover, & Dodd (2005) point out that the “wild-caught” amphibian and reptile trade is largely unregulated and occurs in a void of knowledge about the ability of species to tolerate that collection. They comment,

The management and monitoring of amphibians and reptiles may have been historically overlooked because of a perception that the level of take was insignificant relative to natural rates of replenishment. Now, however, amphibians and reptiles are experiencing global declines, and the commercial trade is a global force that has the potential to contribute to these declines. (Schlaepfer et al., 2005, p. 263)

Parry-Jones and Englar (2007) report that the trade in free-living animals (including for “pets”) is “huge and escalating” and includes rare and endangered species with potentially “devastating” effects. Laidlaw (2005) warns that the exotic “pet” trade (including reptiles) is “driving species toward extinction and disrupting ecosystems” (p. 5).

Collection of “wild” reptiles also frequently involves practices that are ecologically damaging—for example, targeting mature and thus reproductively important individuals and the incidental take of particularly vulnerable and protected species (Warwick, 1997; Laidlaw, 2005). A 2013 study by 200 scientists led by the Zoological Society of London found that 19% of the world’s reptiles face the threat of extinction, with deforestation and biological harvesting—including for the meat and “pet” trades—cited as relevant causes (Bohm et al., 2013).

Contrary to some beliefs, captive breeding is often not a solution to species conservation and ecological problems, and the perception that captive breeding is curative of the problems associated with “wild-caught” reptiles may actually increase some “wild-caught” animal trade (Warwick, 1997; Akhtar, 2012). To begin with, many claims that animals are captive-bred are actually unverifiable or false. TRAFFIC (2012) has reported that tortoises, turtles, birds, frogs, lizards, snakes, and mammals are notable components of false “captive-bred” declarations and has commented, “Most people would never imagine their pet might have been sourced illegally, that they have inadvertently supported wildlife crime, or that their purchase could have contributed to the threatened status of species in the wild.” Next, it is often difficult to distinguish between “wild-caught” and captive-bred animals (Akhtar, 2012), and many so-called captive-breeding operations supplement both their breeding stock and their sales stock with “wild-caught” animals (Akhtar, 2012; TRAFFIC, 2012). In addition, even where genuine, captive breeding frequently presents a groove through which “wild-caught” animals are concomitantly traded (Warwick, 1997; Akhtar, 2012). Accordingly, “wild-caught,” captive-bred, legal, and illegal animals are all traded under the façade of “captive-bred.”

Human population has increased from 1 billion in 1900 to 6.5 billion today (Chomel et al., 2007), and it is predicted that by 2032 more than 70% of the land surface globally will have been damaged or disturbed (UNEP, 2002). Habitat loss has been identified as a major threat to 85% of all species described in the International Union for Conservation of Nature (IUCN) Red List (those species classified as threatened or endangered; WWF, 2011). In short, there are simply fewer free-living animals out there, making the human impacts on free-living animals now greater than ever. It is not difficult to argue that the exotic “pet” industry is one unnecessary burden the world does not need.

Invasive Alien Species

Escaped or unwanted and released “pet” reptiles have become a significant concern in numerous countries because of the actual and potential consequences of these animals forming introduced or “invasive alien” species (Arena, Steedman, & Warwick, 2012; ENDCAP, 2012; Toland et al., 2012). Invasive alien species are those animals and plants that once in foreign habitat essentially become a “pest” by altering indigenous ecology—

for example, through competing with less hardy local “wildlife.” Along with climate change, overexploitation, pollution, and habitat alteration, invasive alien species (IAS) are regarded as one of the major threats to the natural biodiversity of Europe (Shine et al., 2010). Unnatural competition and predation of indigenous animals by introduced species, contamination from novel pathogens, and alteration of ecosystems are all major concerns. IAS can also impact on human health—for example, through injuries and poisonings from lizard and snakebites (NISC, 2013). IAS-related problems cost hundreds of billions of dollars in economic damage and ecosystem management programs around the world (Strayer, Eviner, Jeschke, & Pace, 2006).

Although the proportion of IAS damage attributable to released reptile “pets” is unknown, the effects are believed to be significant (ENDCAP, 2012) and probably increasing along with wildlife trade. Of the 174 European species listed as critically endangered by the IUCN Red List, 65 are in danger as a result of invasive species (EEA, 2010). Fifty amphibian and reptile species have become established in Europe (Kark et al., 2009), and a recent survey indicated that approximately 51 types of nonnative amphibians and reptiles live “wild” in the London, England, area alone (Langton et al., 2011). Taking reptiles from where they do belong, or breeding them, and putting them where they do not belong appears to be a double-edged sword with no blunt end.

Do Trade and Keeping Generate Conservationists?

Both the pro-reptile “pet” community and some well-regarded biologists believe that reptile keeping often engenders respect for free-living animals and constitutes a formative period in the development of future conservationists. Here, the implication is that regardless of the fate of any animals, the ends justify the means for how and why some conservationists do what they do. This perspective might be modestly persuasive if there existed the evidential gravitas to support it. For that to be the case, several things (at least) would need to be established.

Reptile trading and keeping would have to produce either many active, dedicated, and—above all—effective conservationists or a few extraordinary individuals whose endeavors outweigh the indisputable harm surrounding practices involving captive reptiles. It is worth recapping just a few sample areas of concern. In the United Kingdom alone, annually at least 700,000 (Toland et al., 2012) “wild-caught” and captive-bred reptiles enter the reptile trade and keeping “system,” and four out of five die prematurely in their first year (Toland et al., 2012). Invasive alien species continue to dominate many world conservation headlines, with released “pet” reptiles being an important component of that threat. Claims that the selling and keeping of reptiles aids species conservation reside between speculation and falsehood. What is well established is that trading and keeping reptiles (both “wild-caught” and captive-bred) can directly and indirectly harm species populations, whole species and entire ecologies (Akhtar, 2012). Incidentally, even where some effective conservationists emerge who attribute their role to earlier reptile keeping, can it truly be said that they would not have chosen that path anyway?

Regardless, even if the argument that the ends justify the means was considered to be ethically acceptable, the reality is that the collateral damage from reptile trading and keeping is expanding, not receding. Accordingly, there does not exist a genuinely significant impact from ambassadorial conservationists who were once reptile keepers, and the "philosophy," if it ever possessed a sound rationale, has catastrophically failed.

COMMERCE AND ECONOMICS

Industrial Freedoms or Liberties?

The commercial and economic dynamics of reptile trading share many commonalities with other industries, but also several significant differences. Many if not most of the things we engage with have been rigorously and independently tested prior to being made publically available and are subsequently controlled by stringent regulation. They are typically "pre-checked" regarding their safety for people and the environment rather than being openly marketed and retrospectively investigated or controlled. The trade in reptiles and "wildlife" in general is one of few industries anywhere in the world that escapes entirely these protection norms. In fact, in almost all countries the monitoring process for the entire trade operates in reverse of this precautionary approach.

Essentially, there are two systems used to "control" such trade in animals. The standard and almost universal system is the "negative list." This is the fundamental structure of the Convention on International Trade in Endangered Species (CITES) and involves placing species on monitored, restricted, or banned lists. Species are added only following an inertia-laden and trade-favoring process that applies to a minority of species, and the system is poorly regulated (Warwick, 1997; Laidlaw, 2005). The alternative system used by a few countries is the "positive list" and was historically also known as the "green list" or "reverse list." The positive list turns the standard negative list on its head and implies that no species can be traded unless it is first proven safe for commerce under independent scientific evidence-based criteria. A "positive" listing does not automatically suggest that something is good for people, animals, or environment, but merely indicates that it has been checked to establish that it is not people- and environment-unsafe. Where the system is used, it is significantly superior to the negative-list principle of CITES. The positive-list system uses the same principle as virtually all other industries. Trains, planes, cars, buildings, electrical goods, drugs, and even people—dentists, doctors, pilots, almost anyone whom the public may rely on for their well-being—are subject to a list of verified and approved (effectively "positive-listed") things or professionals.

Unlike most producers, there are no requirements for reptile traders to prove that their "merchandise" is safe for the public, safe for the environment, or even safe for the so-called product (a live animal) itself. In fact, unlike other industries, "wildlife" trading is systematically given the benefit of the doubt—meaning that trade continues in any direction and with any implication until others, in their own time and at their own cost, gather all salient evidence and gain protective legislation. Even where successful, all this amounts

to is certain species being included on negative lists (i.e., animals who are restricted or banned), and traders and keepers then commonly either flout the regulations or move on to another species, starting the entire cycle again. So while the “wildlife” industry steams ahead almost unencumbered, animal welfare, species conservation, environmental, and public health organizations and campaigners labor against governmental regulatory inertia wherein the “control” of practices amounts to little more than a fluctuating smokescreen.

If the normal protections required of responsible industries were directed at the trade of reptiles (and other animals), then the business would comprehensively fail on all counts. For example, in terms of animal welfare, the “products”—sensitive, sentient creatures—will probably die (fail) within a year, making neither “manufacturer’s” nor statutory guarantees possible. Regarding public health and safety, all reptiles would fail hygiene tests at the very least. In terms of environmental impact, between damaging collection methods and release of invasive (arguably even toxic) species into nature, reptiles would also fail a number of these precautions.

A simple example of how the reptile trade fails its customers where other industries at least aspire to succeed can be represented using a child’s toy—for convenience we can make that example a soft fabric turtle. To be sold, the toy turtle will typically require eyes that cannot be pulled from the product, so that the child cannot ingest or choke on them. There will be no sharp edges that could cause any incidental harm. The materials will be environment- and child-safe, in case the child puts the toy to his or her mouth. Also, the whole thing will be machine-washable so that it does not become an infection hazard. There is a reasonable chance the toy will also be fire-resistant as an extra safeguard. And finally, a few stitched-in labels will demonstrate the hoops that had to be jumped through to get that toy turtle onto the market and who behind it is fully accountable for any failures.

Compare that with the reptile trade, where anyone can stroll into their local animal store, demand a live turtle, and deliver that animal right into the hands of a child. Unlike the soft toy, surface contaminants, skin detritus, feces, and bacteria likely reside on and in the turtle. Sharp jaw plates and claws can easily cause injuries and result in infections. Turtles cannot be cleaned of their “contaminants”—typically these are as normal to the turtle as our own bacteria are to us. In brief, there can be no genuine guarantees whatsoever about either the safety of the “product” or the turtle’s own health, and a purchaser is unlikely to know where the animal came from, how the animal got here, and what to do with him or her. All this raises serious questions about people’s capacity to practice informed consent when purchasing a reptile. Worse, in contrast to the dearth of information about any aspect of live turtle safety, there exists a veritable library of independent, scientific, and medical material that unequivocally demonstrates that “pet” turtles, like all reptiles, are distinctly unsafe for children, as well as for many normal healthy adults (Mermin et al., 2004; Chomel et al., 2007; Brugere-Picoux & Chomel, 2009; Smith et al., 2012; Warwick, Arena, Steedman, & Jessop, 2012).

The notion of having to conform to the normal requirements of the positive list system strikes fear deep into the hearts of reptile traders and keepers because it would mean

an end to the multi-decade-long free ride that they have enjoyed while animals, many people, and the environment have not.

Balancing the Books?

Why the reptile trade and other trades in nondomesticated animals have enjoyed this red-tape free ride for decades remains a mystery. Big business is often perceived as getting away with things simply because of its basic economic importance and the financial feedback to the state. The trade in animals is certainly big business. For example, a 2009 TRAFFIC report estimated that all legal trade in “wildlife products” was worth approximately €100 billion for the European Union in 2009 (ENDCAP, 2012). However, this would include skins and furs as well as some live animals for “pets.” A 2007 report cited a value of €7 million for live reptiles, and €100 million for skin-based products (Engler & Parry-Jones, 2007). Generally, captive-bred animals within the European Union and United Kingdom will not show in these data, so these figures understate the scale. But do these numbers add up to justify the incidental exemption of reptile trade from commercial norms? Representative data specifically for reptiles are lacking. However, taking “wildlife” trade as a whole, the €100 billion figure sounds impressive until one considers at least some of the covert costs and destructive disbursements.

In the European Union the annual cost of damage from IAS is estimated at approximately €12.5 billion—many of these invasives are released “pet” reptiles (European Commission, 2011). That is only part of the picture. There are over 10,000 animal and plant species alien to Europe on the continent now, yet ecological impacts are documented for only 11% of the total and economic impacts for only 13% of species (Vilà et al., 2010). The costs in global biodiversity loss from reptile capture for “pets” are not known. However, a recent report concluded that protecting even the world’s most endangered species from extinction would cost over \$76 billion a year (McCarthy et al., 2012). There are also public health costs and costs of health care for treating reptile-borne human disease: a single doctor’s consultation can be €50, a single hospital consultation €250, and a single day in the hospital €2,500. To this one can add days off work and lost productivity.

What broader costs are associated with ecological damage from reptile “pet” trading? One can add a variety of costs, including public money donated to charitable causes (animal welfare, species conservation, and public health) that work on relevant problems; costs of managing the array of largely ineffective but expensive “regulatory” organizations; and local, national, and international governmental and parliamentary time spent on reviews and legislation, however ineffective. Across a nation, these costs will be high—across the European Union, very high. It is highly questionable whether, even on economic grounds, the reptile trade can balance its own books.

CONCLUSION

Reptiles are not afforded all the same considerations as traditionally “cute” animals. At least three reasons probably account for this. First, the anthropomorphic bond is

dramatically weaker with reptiles than with many other animals. This non-communication may be associated with or result from certain typical reptilian anatomical features, such as minimal facial expression and vocalization. Second, reptile behavior is often not socially centered, making reptiles less “interactive.” Third, awareness within both the scientific and lay communities regarding indicators of stress and disease in reptiles is poor, and thus signs (and suffering) that may be obvious to experienced observers can typically go unrecognized by others.

Would dog caregivers put their loved companion in a vivarium and observe him or her clawing at the glass for hours each day? Would society accept the vast majority of dogs dying within one year from mistreatment, maladaptation, and stress-related disease? The answer, one hopes, in all these cases has to be a resounding no! A major result of this reptilian–mammalian mismatch is that reptiles are routinely subject to abuses by people who probably would not direct the same treatment at familiar and intrinsically valued animals.

It is difficult to avoid the conclusion that reptile trading and keeping operates largely on a mis-evolved and maladapted belief system based on outdated biological understanding and cumulative erroneous presumptions. The harm and failures endemically caused by those who sell and keep reptiles are compounded by regulatory inertia among those with whom resides the formal responsibility to prevent and control the negative consequences of the entire “business.” There are no valid reasons that reptile trading and keeping should avoid the normal precautionary investigation and control (e.g., the “positive list”) required of other industries and, arguably, no good reason that reptiles should be sold or kept as “pets” at all.

One inarguable fact is that people benefit financially or derive other satisfaction from trading and keeping reptiles. Another inarguable fact is that countless individual animals, species, environments, ecologies, and people suffer to produce those benefits.

Reptiles are physically, emotionally, and psychologically sensitive. They possess complex biological needs, and their sensitivity and sentience is clear to anyone who chooses to be mindful of these animals. Few people who hold them captive seem to appreciate those needs, or look beyond the mundane practicalities of how one can obtain and “keep” a reptile, to whether it is right to do so. Perhaps ironically, although reptiles possess the cold-blooded reputation, it is humans themselves who are emotionally insensitive—to the detriment of reptiles, humans, and the world in which they live.

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